

BioBrew

University of San Diego, Senior Design

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Did you know?

~310,000,000

0

metric tons

of CO₂ are emitted per year due to the
life cycle of alcoholic beverages

Motivations for our Design



Environmental Impact

Less CO₂ released into the air through natural algae sequestration.



At Home Replication

Ease of setup so that homebrewers can use this system at home

Problem Statement

The goal of this project is to **demonstrate a sustainable method of reducing CO₂ emissions** generated during beer fermentation. We will do this by developing a **system that will direct the CO₂ concentrated off-gas** from homebrewing fermentation into an **algae subsystem which will absorb CO₂ via photosynthesis**. This system will be replicable by homebrewers and is potentially scalable to larger-scaled craft brewing operations.

Product Requirements



Direct

Feed gas within
the system



Maintain

Algal health



Connect

With home brewing
fermentation set-up



Reduce

CO₂ based on feed
gas demand



Prevent

Algae and CO₂
backflow into beer

Methods of Verification

Spectrophotometry

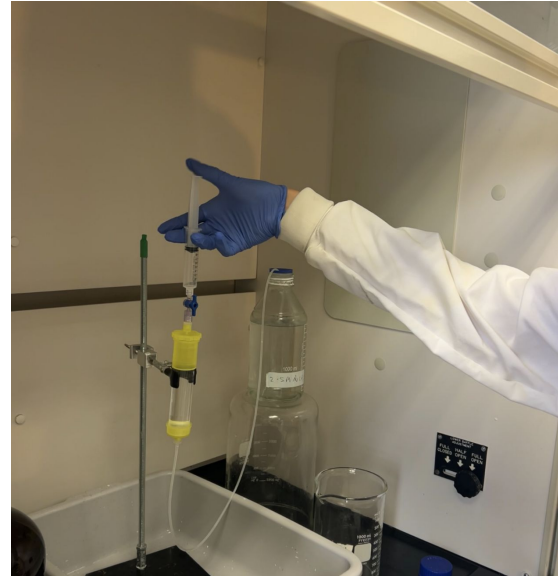
To measure algal health



Cuvette

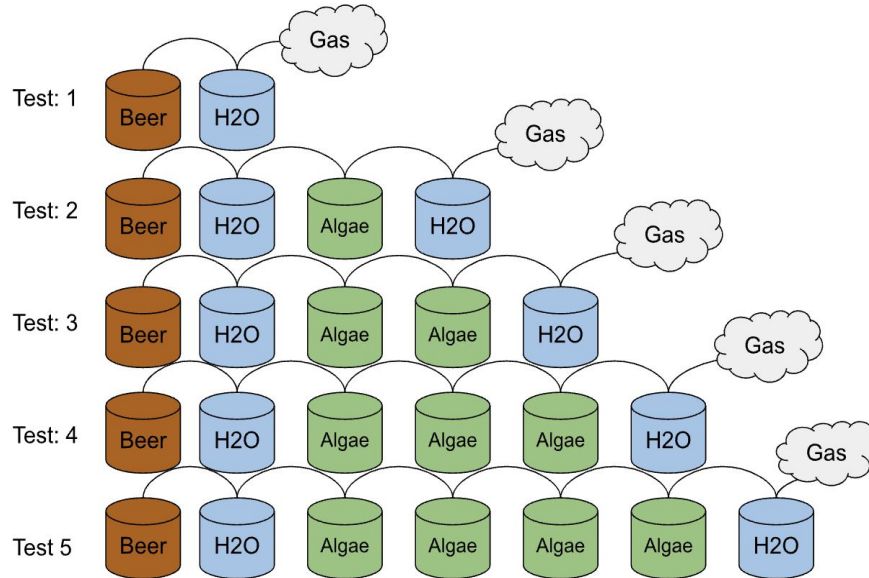
NaOH Scrubbing

To measure CO₂ concentration



Algae-to-Beer Critical Function Prototype

Test plan:



Actual setup with gas capture:

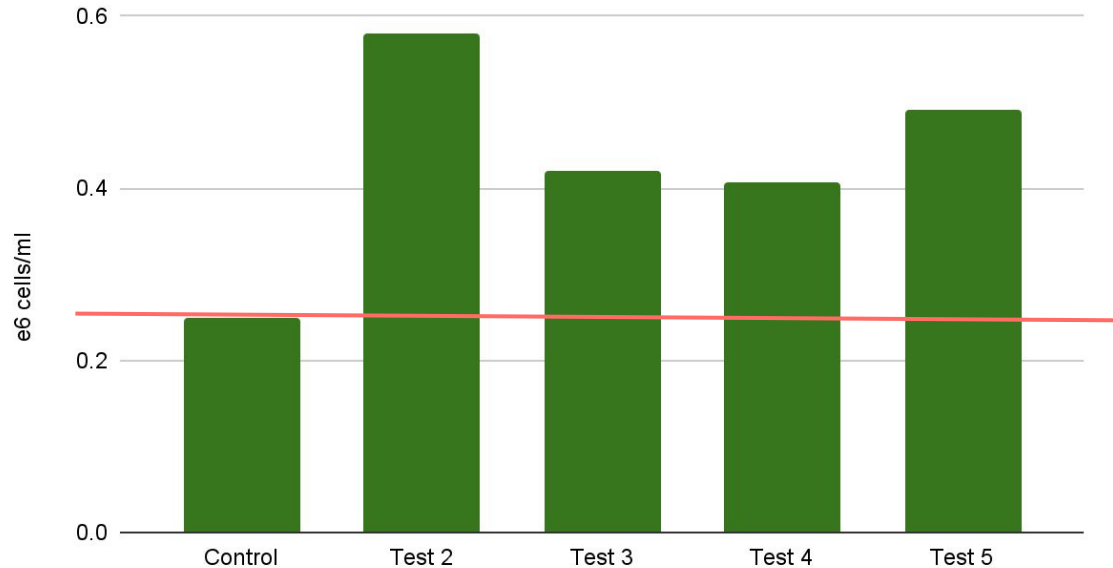


Test Results: Algal Health Testing

4 different volumes of algae:

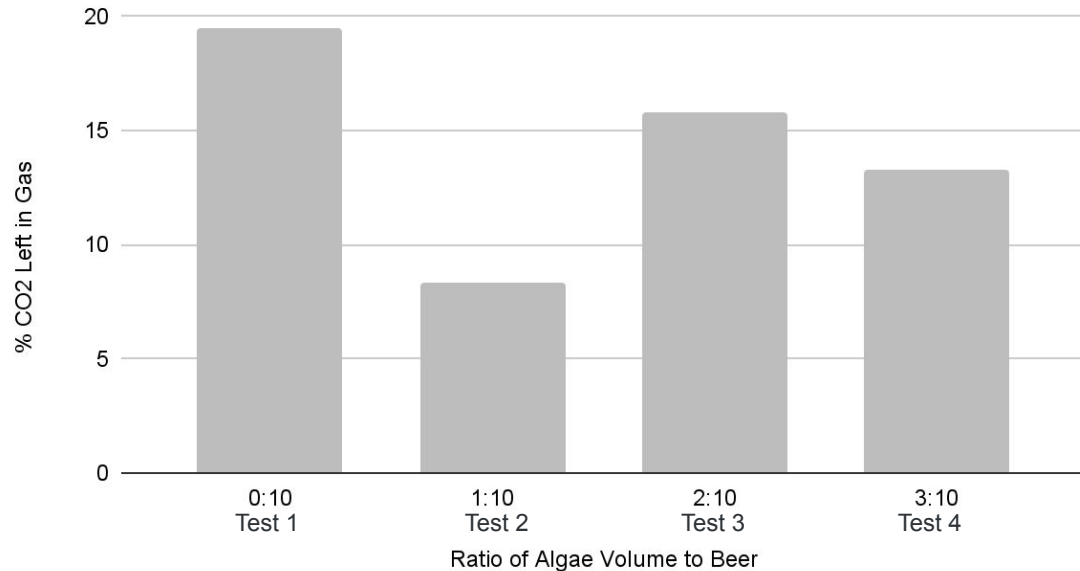
- **Test 1:** no algae present
- **Test 2:** 40mL
- **Test 3:** 80mL
- **Test 4:** 120 mL

5-Stage Algal Health Test Results



Test Results: CO2 Concentration Testing

Concentration of CO2 Post Algae System



4 different volumes of algae:

- **Test 1 (control):** no algae present
- **Test 2:** 40mL
- **Test 3:** 80mL
- **Test 4:** 120 mL

Algae Specifications

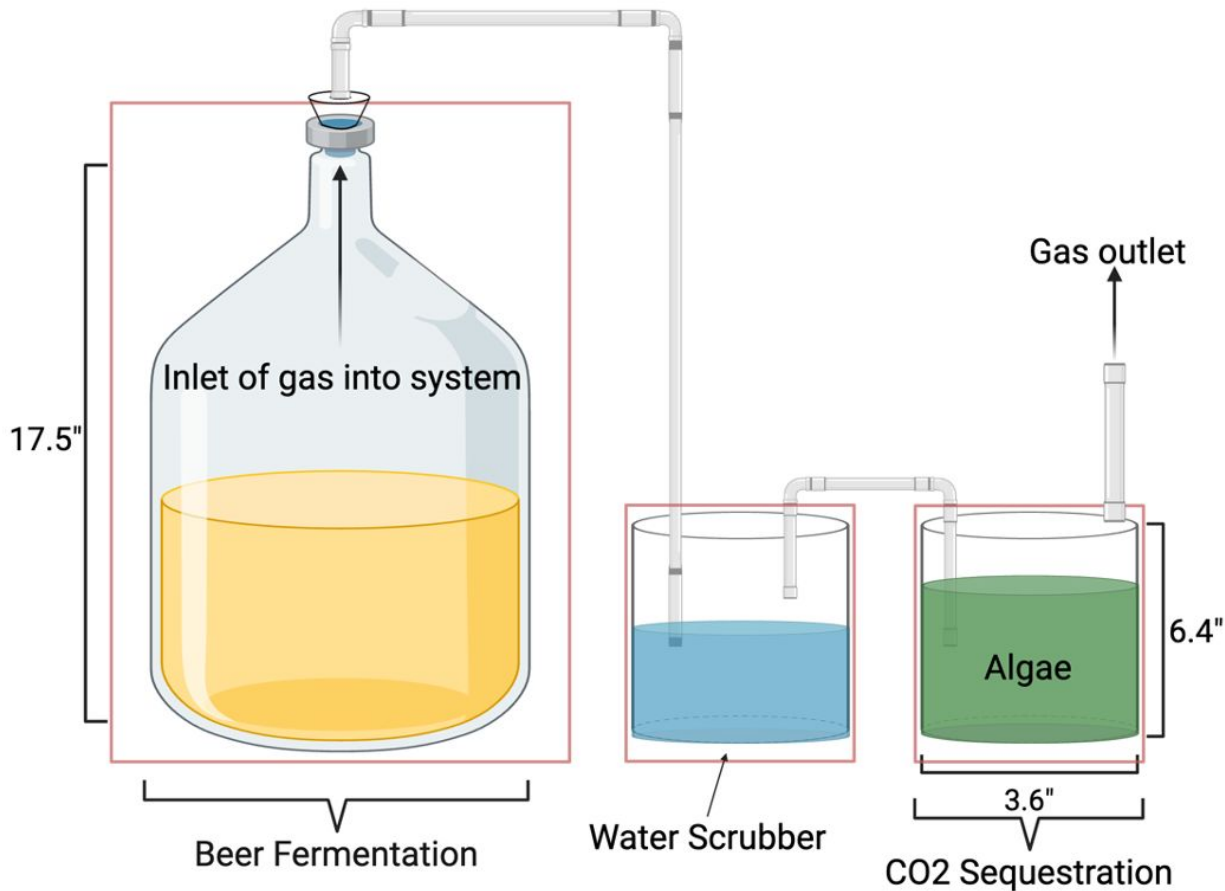
From our design analysis testing we concluded that **one algae container** was optimal for algae health and that a **ratio of around 1:10** for algae to beer allowed for the most significant reduction in CO₂.

Final Design:

One algae container with a volume ratio of 1.32:10 to reach an even volume of 500 mL to facilitate ease of setup in home brewing application.



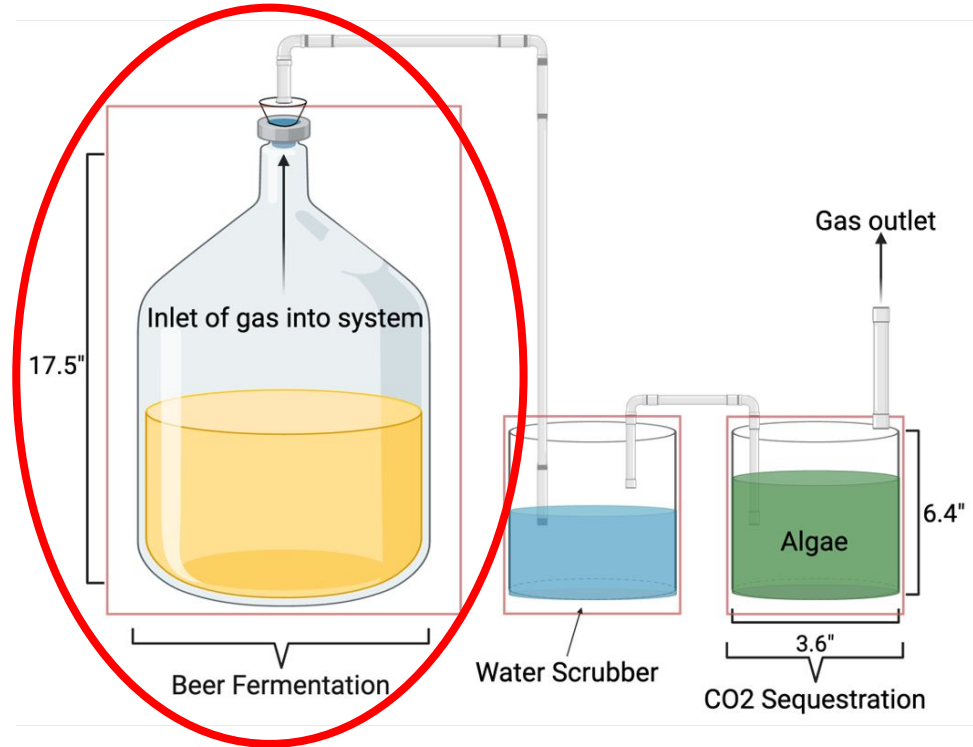
Final Design



Subsystem Specification

Fermentor

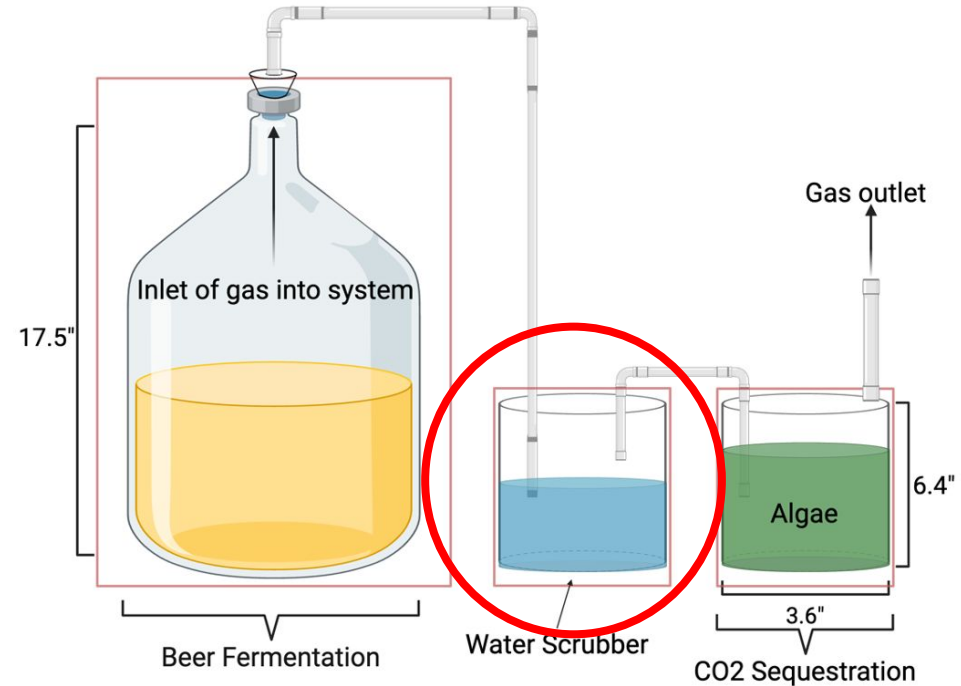
- Produces gasses from fermentation
- Source of CO₂ production



Subsystem Specification

Water Separator & Scrubber:

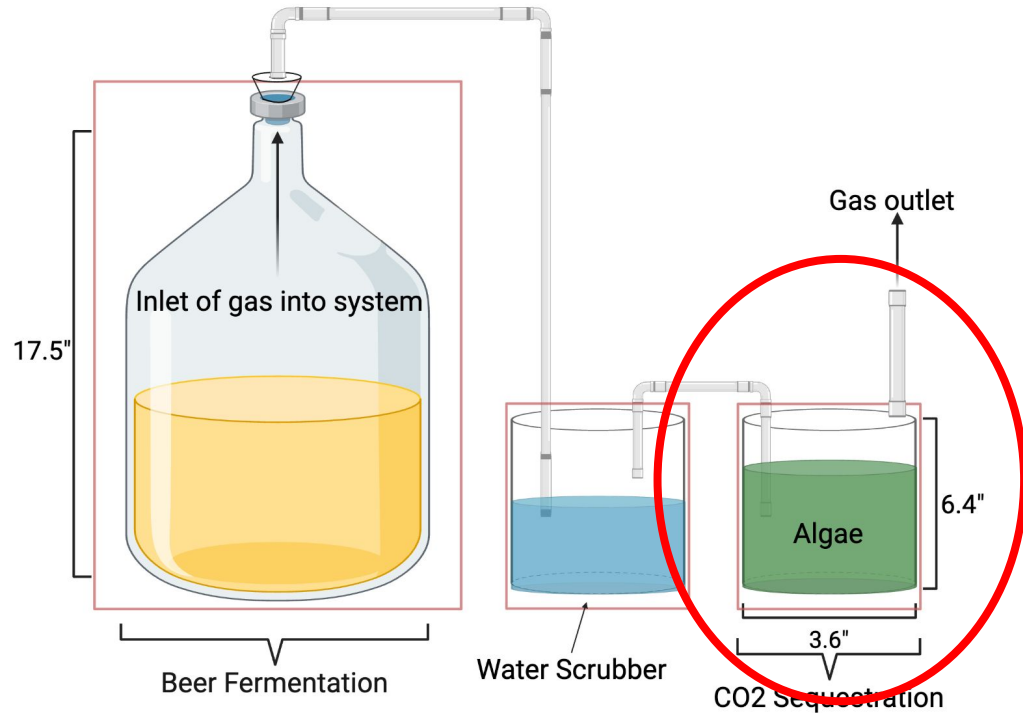
- Dissolves and catches any waste
- Prevents contamination of beer
- Prevents contamination of algae from beer waste



Subsystem Specification

CO₂ Sequestration:

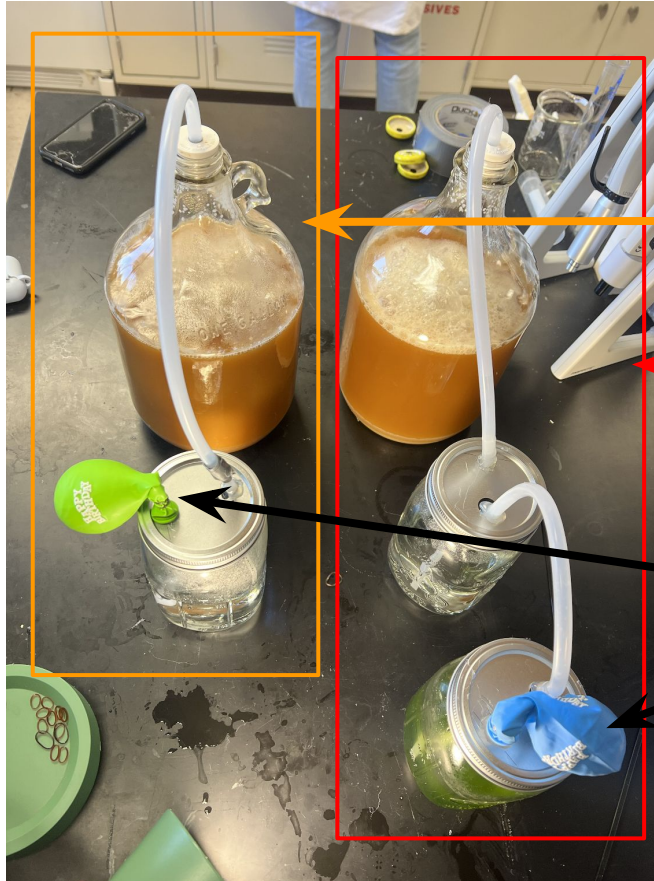
- 500 mL algae
- Algae captures CO₂ and produces oxygen through the process of photosynthesis
- Reduce CO₂ at output



Implementation: Final Design



Implementation: Device

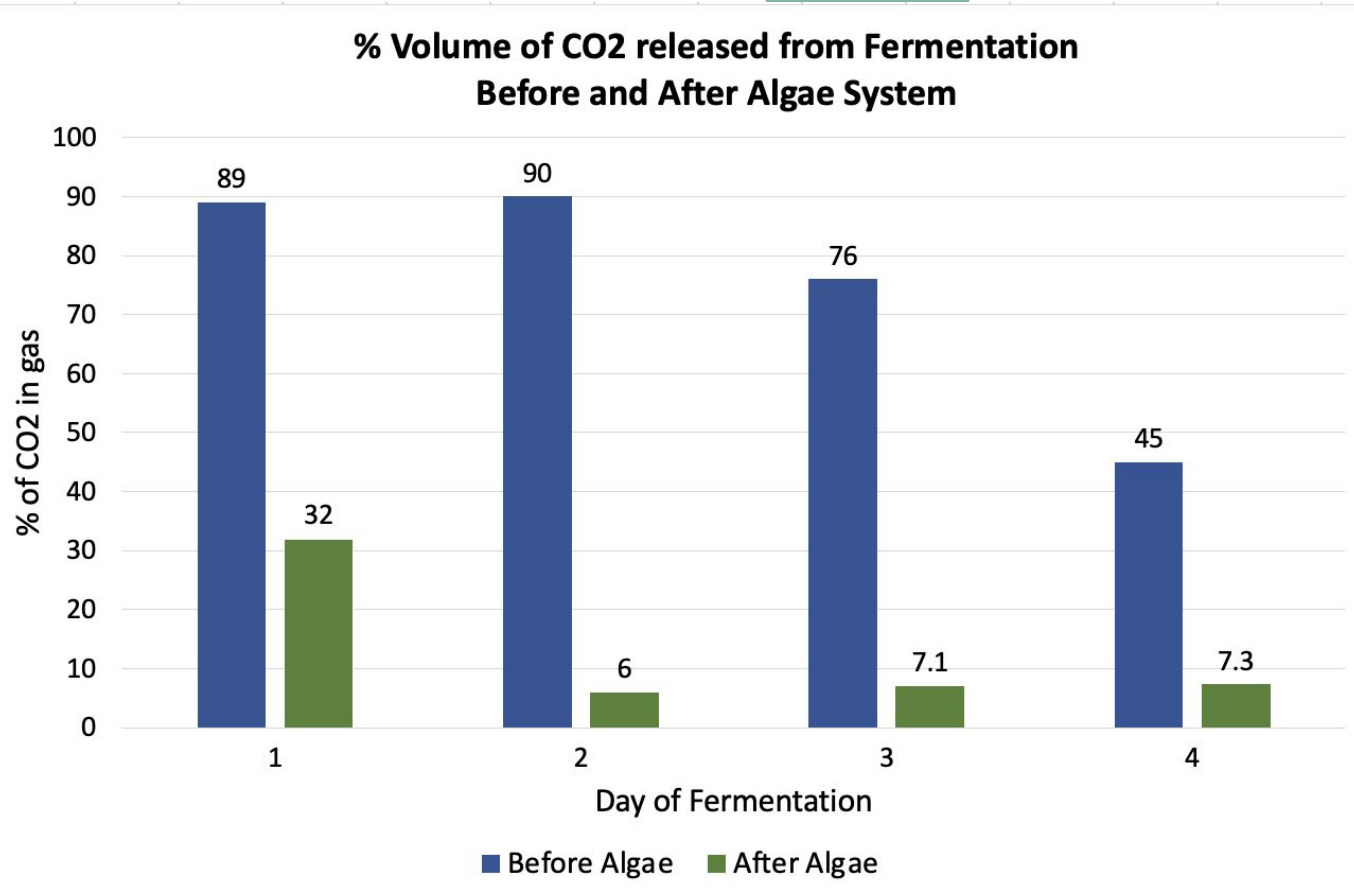


No algae present

Our System with algae

Balloons to capture &
test CO₂ concentration

System Validation: CO₂ Concentration Testing



Home Brewing Blog

Want to know more?

Read our blog post to learn more about:

- Step-by-step instructions for system setup
- An Excel sheet linking all materials
- Algae cultivation suggestions & resources



FINAL DELIVERABLES

01

Successful Method of Capturing CO₂

Determined a successful method to capture CO₂ from home brewing setup

02

Setup & Run Full Fermentation Cycle

Setup home brewing system attached to algae and run full fermentation cycle

03

Prove Reduced CO₂

Proved design to be reducing CO₂ output via NaOH Scrubbing

04

DIY Instructions

Created a blog post with instructions for DIY replication of setup for homebrewers

Questions?



Timeline

FALL SEMESTER

September:

- problem statement, requirements, research
- CB:
 - assigned project to clean CO₂ using an EDBM
 - no more EDBM but use a CO₂ scrubber and carbon filter

October:

- initial CFP ideation, CDR Presentation, CFP build & test plan report
- CB:
 - Meeting with Mach and informed not to use a scrubbing tower

November:

- CFP Testing, feasibility analysis

December:

- PDR, fall engineering expo

SPRING SEMESTER

February:

- CB: drops project
- BioBrew is born, new problem statement, requirements, research

March:

- CFP ideation, Feasibility analysis, preliminary budget submittal, CO₂ sensor research & troubleshooting, CDR presentation

April:

- CO₂ sensor troubleshooting, Build & Test Plan, CFP testing, CO₂ sensor housing 3D printing, started spectrophotometer readings

May:

- Scrubbing setup, NaOH scrubbing, final design, more CFP testing, ran multiple fermentation cycles, blog post, FDR, spring engineering expo

Budget

Quantity	Item Description	Subsystem	Part Number	Vendor	Link	Unit Price	Cost	Notes
3	Wide Mouth Mason Jars [32 oz]	CO2 Sequestration H2O Scrubber	B084Q7FQWH	Amazon	Amazon: Mason Jars	\$5.00	\$15.00	6 pack - (\$29.99 total)
2	Algae Culture Kit for Chlorella Vulgaris [3.8L]	CO2 Sequestration	AKCv-03800	Algae Research & Supply	Algae Research & Supply: Cu	\$32.99	\$65.98	Includes algae and nutrient media
7	Rubber stopper w/ 3/16" hole	Beer Fermentation	B06XSKWKBG	Amazon	Amazon: Rubber Stopper	\$1.23	\$8.61	10 pack - (\$12.31)
2	CO2 Sensor	CO2 Detection	2091-SPRINTIR-W-F-100-ND	Digikey	Digikey: SPRINTIR-W-F-100-N	\$236.45	\$472.90	Notes
1	Plastic Tubing	General Assembly	B08TKXPHX5	Amazon	Silicon Plastic Tubing	\$9.59	\$9.59	10 foot length
5	Calypso Lemonade Glass Bottle [16oz]	Beer Fermentation	B0852SLSRF	Amazon	Calypso Glass Bottles	\$4.08	\$20.40	6 pack (\$24.50 total)
1	Algae Culture Kit for Spirulina [3.8L]	CO2 Sequestration	AKAp-03800	Algae Research & Supply	Algae Research & Supply: Spi	\$32.99	\$32.99	Includes algae and nutrient media
3	Glass Beer Fermentation Jug	Beer Fermentation	B006408Z76	Amazon	Beer Fermentation Jug	\$16.85	\$50.55	Can also be bought at brew marts. One broke during testing.
8	CBW Dry Malt Extract - Sparkling Amber [1 lb]	Beer Fermentation	B0100T822Y	Amazon	Fermentation Wart/Malt	\$10.74	\$85.92	Mostly bought via personal purchase
1	Gas Flow Meter 1-10LPM	General Assembly	B07DNW6JHX	Amazon	Gas Flow Meter 1-10LPM	\$20.53	\$20.53	Notes
1	CO2 Sensor	CO2 Detection	2091-EXPLORIR-M-E-100-ND	Digikey	Digikey: EXPLORIR-M-E-100	\$185.50	\$185.50	Notes

Total budget: \$1500

Total: \$967.97

See BOM & Total Budget Here: [Link](#)

CO₂ is Produced & Used in the Brewing Process

Yeast + Sugar

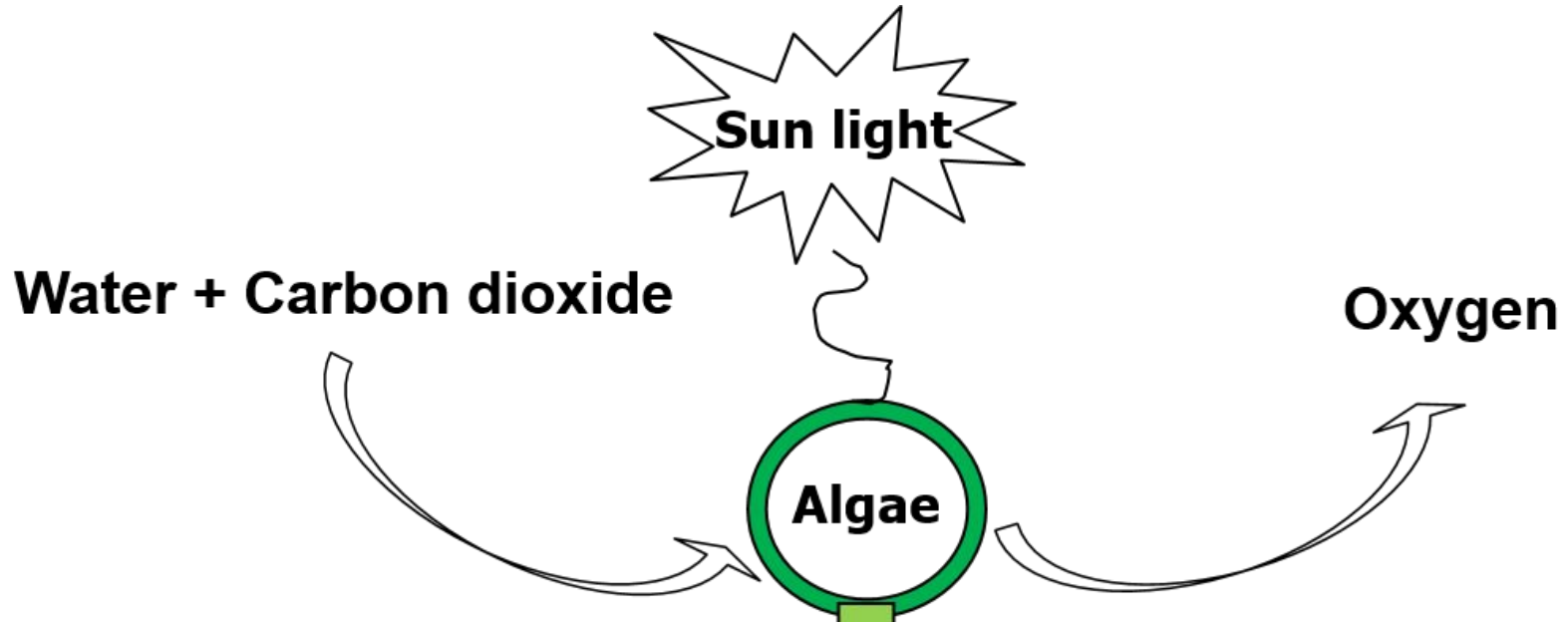
Fermentation



CO₂ Molecules

Alcohol

Algae Photosynthesis can Reduce CO₂



How much algae would you need to upscale to a craft brewery?

Fermenter = 50 BBL = 315 gallons

so

41 gallons of Algae for a craft brewery fermenter